Bhargav M

+91 7795036940 | bhargav.m23@iiits.in | LinkedIn | GitHub

Summary

B.Tech Artificial Intelligence and Data Science student graduating in 2027, with projects in image classification, deep learning, geospatial data processing, and data validation. Uses Python, SQL, and image-processing tools to build and evaluate data-driven applications. Seeking a **Data Scientist** internship at **Vassar Labs**.

Work Experience

CYMAX Jan–May 2026

Unity Developer Intern

- Developed a Unity 360-degree video player for VR headsets with spherical rendering, headset/controller input, and playback controls.
- Built Python services and a React playback client for VR streaming, integrating authentication, token-controlled access, and encrypted video delivery.

Indian Institute of Information Technology, Sri City May–Jul 2026

XR and Simulation Intern

- Developed the Alipiri–Tirumala AR Navigation app as part of a collaborative project with **IIT Tirupati** and **Tirumala Tirupati Devasthanams (TTD)**.
- Processed GeoJSON route segments and landmark coordinates; combined GPS with visual-inertial localization for offline walking guidance.
- Developed a demo environment for ISRO's third launch pad using **Unreal Engine 5**.

Projects

OpthaMiss | Deep Learning for Eye Screening GitHub

Python, PyTorch, Vision Transformers, Pandas, NumPy, Scikit-learn, OpenCV

- Developed image-screening pipelines for anterior and fundus eye images with glare removal, black-border removal, CLAHE enhancement, and test-time augmentation.
- Implemented a Vision Transformer training workflow with progressive unfreezing and class-balanced sampling; added ROC-AUC, F1, precision, recall, and confusion-matrix evaluation.

Plant Disease Detection | Agricultural Image Classification GitHub

Python, TensorFlow, Keras, NumPy, OpenCV, scikit-image, Flask

- Built a leaf-image classification application using TensorFlow/Keras models, with preprocessing for segmentation, background removal, contrast enhancement, and noise reduction.
- Added Gabor texture features and preprocessing visualizations; returned ranked disease predictions and confidence scores through a Flask interface.

Alipiri AR Navigation | Geospatial Data Processing GitHub

Unity, C#, GeoJSON, GPS, Kalman Filtering, A Pathfinding*

- Chained route segments by endpoint proximity, corrected segment direction, and validated landmark coordinates; flagged a 1.17 km source-data gap as interpolated guidance.
- Implemented GPS filtering, route snapping, and A* routing; built a navigation HUD to display route progress, ETA, and sensor/tracking status.

Excel-to-App | Calculation and Data Validation GitHub

Python, SQL, FastAPI, PostgreSQL, SQLAlchemy, pytest, Docker

- Built a spreadsheet formula parser, dependency graph, cycle detection, and a 66-function evaluator with versioned APIs and execution audit records.
- Validated calculations against LibreOffice, documenting **481/481 matching comparisons across 37 cases** on a shipping-calculator fixture.

Technical Skills

Programming and Data: Python, SQL, C#, Pandas, NumPy, PostgreSQL, SQLite
ML and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, Vision Transformers
Processing and Evaluation: OpenCV, scikit-image, image preprocessing, augmentation, ROC-AUC, F1
Visualization and Tools: Matplotlib, Seaborn, GeoJSON, Git, Docker, pytest, FastAPI

Education

Indian Institute of Information Technology, Sri City 2023–2027 (expected)

B.Tech in Artificial Intelligence and Data Science

Class XII: 92.3% – BGS PU College, Sidlaghatta | Class X: 97.6% – BGS Public School